My 3d printable headphones consist of a headband, yoke arms, driver holders, and rear cavities, you have to print everything twice, except the headband. There are multiple options for almost every part, for optimising the headphones for the different speakers, and different 3d printers.
You will need two headphone speakers, aka drivers, I made more than one pair of headphones, so I tested a few speaker models, like the Dayton Audio CE38MB-32, and CE30MB-16B, these are pretty cheap, and sound decent. But any 3 cm or 4cm drivers are usable, you can also refurbish drivers from old broken headphones. You also need a type of jack-jack cable, which has two parts for the left and right channels, and these can be separated. If it's long enough you can make two pairs of headphones with one cable, by cutting it in the middle. Besides these you will also need some silicone sealant, blu tack, two 90 or 100 mm universal earpads, zip ties, polyester wadding, and some old cotton clothes for padding.
If you use some random 3 or 4 cm drivers, you have to print out multiple pair rear cavities and driver holders with different sizes, and test, which one sounds better. A bigger rear cavity will give a more bassy sound, a smaller one a more precise, but sometimes boxy sound. The deepness of the driver holder means the driver's distance from the ear pad, you have to add the earpad's thickness for this to get the driver-ear distance. A closer driver will have harsher sound, with more treble, a more distant driver will have a more bassy sound.
3d printed parts are not airtight, so you will have to seal some parts with silicone. You have to cover the internal side of the rear cavities, the space between the drivers, and the driver holders, and when you close it permanently the driver holders and the rear cavities also have to be glued together with the silicone sealant. For the testing you can use the blu tac instead of silicone. This sealing is very important, the headphones will sound bad without it. The only situation where sealing is not necessary is, if you make open back headphones, but you need special drivers for that.
Beside the silicone, the driver holders and the rear cavities also have to be hold together by zip ties, they have built in holes for them.
One side of the yoke arms connects to the assembled parts, which consist of the rear cavities and driver holders, the other side fits into the headband.
You can adjust the size of the headphones by rotating the yoke arm 90 degrees, pulling up, or down, and rotating it back.

Optimal parts for Dayton Audio CE38MB-32 (40mm):
- 25mm radius rear cavity
- 17mm deep driverholder
- ~ 13mm deep earpad

Optimal parts for Dayton Audio CE30MB-16B (30mm):
- 20mm radius rear cavity
- 10mm deep driverholder
- ~ 15mm earpad

Printing instructions:
- ► 0.2 mm or 0.3 mm layer height
- headband: 7 wall loops, 30% infill, petg, with supports
- yoke arm: 7 wall loops, 40% infill, petg, with supports (90° threshold angle)
- rearcavity: 4 wall loops, 15%-30% infill, with supports
- driverholder: 4 wall loops, 15%-30% infill, petg